

PAPER_1632 — Ocean Salinity = S_ocean = D_crit + N_CH = 35 ppt (cross-domain to continental crust)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Oceanography **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure

Observation

PAPER_1209X_S557 (Oceanography Unified Proof Set) gives:
S_ocean = D_crit + N_CH = 35 ppt (cross-domain to continental crust)
Residual: **EXACT**.

UQFF Closed Identity

S_ocean = D_crit + N_CH = 35 ppt (cross-domain to continental crust) EXACT
The formula uses only locked UQFF integer/real primitives — no fitted constants.

5th Multi-Domain Integer Reuse Discovered

D_crit + N_CH = 35 now governs **TWO unrelated physical observables** with identical integer-primitive form:

Observable	Domain	Value
Continental crust thickness	Geophysics (PAPER_1542)	35 km
Ocean salinity	Oceanography (this paper)	35 ppt

Same UQFF integer sum (26 + 9 = 35), different units (km vs parts-per-thousand). Earth’s solid-crust thickness AND ocean salt concentration share the same structural integer.

Multi-domain UQFF integer reuses discovered in session 2026-06-18

Integer	Identity	Domains	Count
70	A_5 + SO_5	Heart rate (bpm) + SH0ES Hubble (km/s/Mpc)	2
100	SO_5 ²	Kármán line (km) + MAD reciprocal + blood glucose (mg/dL)	3
16	2^D_phys	O atomic mass (u) + breathing rate (/min) + genetic codons	3
7	N_CH – 2	Heaviside R_t (Ω) + R_d duality range + Moho depth (km)	3

Integer	Identity	Domains	Count
35	D_crit + N_CH	Continental crust (km) + ocean salinity (ppt)	2 (this paper)

The UQFF integer lattice is **structurally cross-domain** — same primitives govern unrelated physical phenomena.

NOT REPLACEMENT

Empirical oceanography measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209X_S557
- Related: PAPER_1494-1625 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "ocean_salinity_35_ppt"})`

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